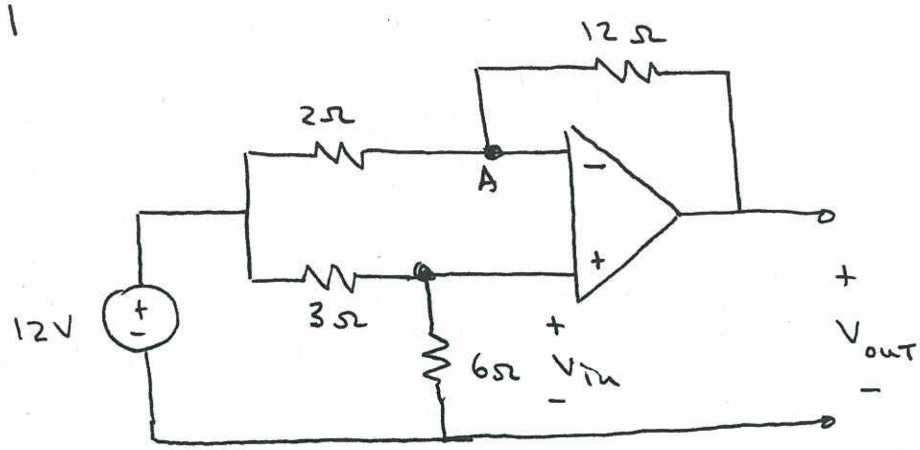
5.1



$$V_{in} = \left(\frac{6\Omega}{3\Omega + 6\Omega} \right) 12V = 8V$$

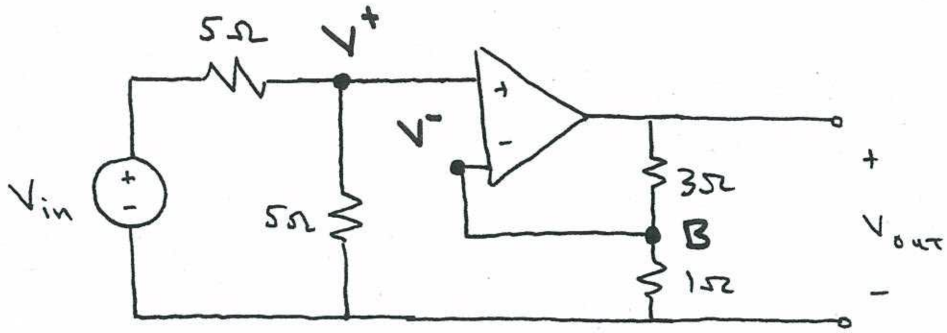
Op-amp rules: $V_A = V_{in} = 8V$

KCL at node A:

$$\frac{12V - V_{in}}{2\Omega} = \frac{V_{in} - V_{OUT}}{12\Omega} \Rightarrow \frac{12V - 8V}{2\Omega} = \frac{8V - V_{OUT}}{12\Omega}$$

$$\boxed{V_{OUT} = -16V}$$

5.2



$$V^+ = \left(\frac{5\Omega}{5\Omega + 5\Omega}\right) V_{in} = \frac{V_{in}}{2}$$

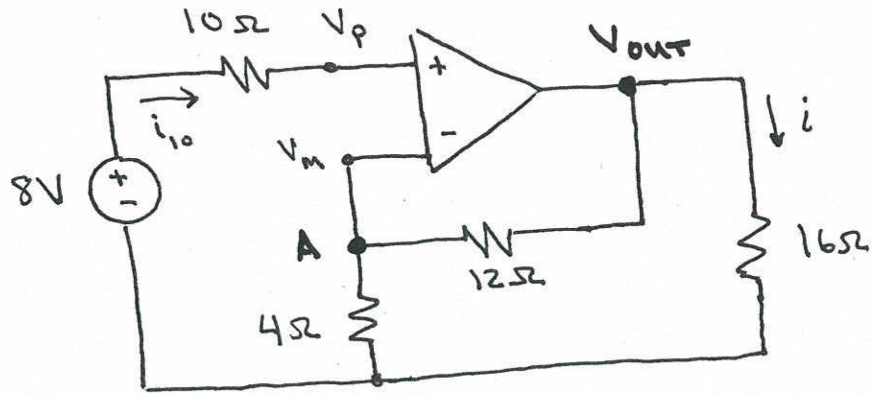
Op-amp rules: $V^- = V^+ = \frac{V_{in}}{2}$

KCL at B:

$$\frac{V_B}{1\Omega} = \frac{V_{OUT} - V_B}{3\Omega} \Rightarrow \frac{V_{in}}{2} = \frac{V_{OUT} - \frac{V_{in}}{2}}{3}$$

$$\boxed{V_{OUT} = 2\,V_{in}}$$

5.3



Apply op-amp rules:

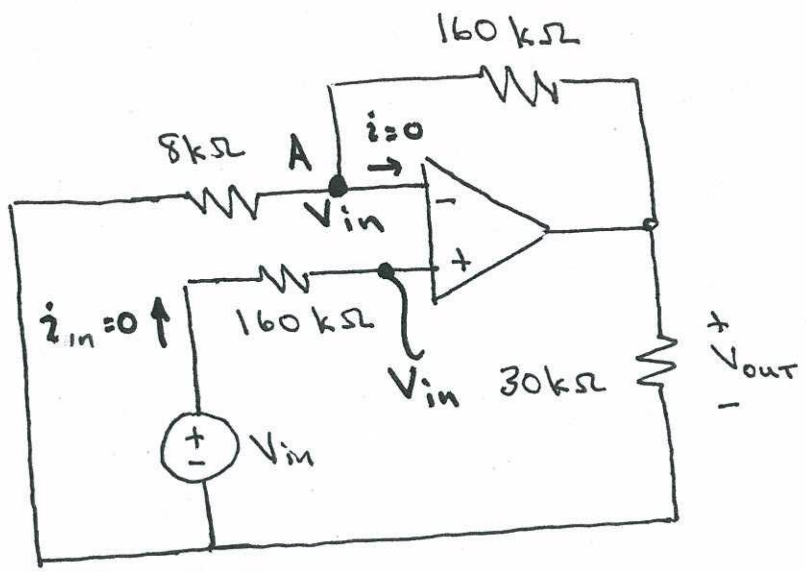
$$i_{10} = 0 \Rightarrow V_p = 8\,V$$

$$V_p = V_M \Rightarrow V_M = 8\,V$$

KCL at "A": $\dfrac{8V}{4\Omega} = \dfrac{V_{OUT} - 8V}{12\Omega} \Rightarrow V_{OUT} = 32\,V$

Ohm's Law: $i = \dfrac{V_{OUT}}{16\Omega} = \dfrac{32V}{16\Omega} \Rightarrow \boxed{i = 2\,A}$

5.4



Op-Amp rules:

$i_{in} = 0 \Rightarrow V_{in} = V_p$

$V_p = V_N = V_{in}$

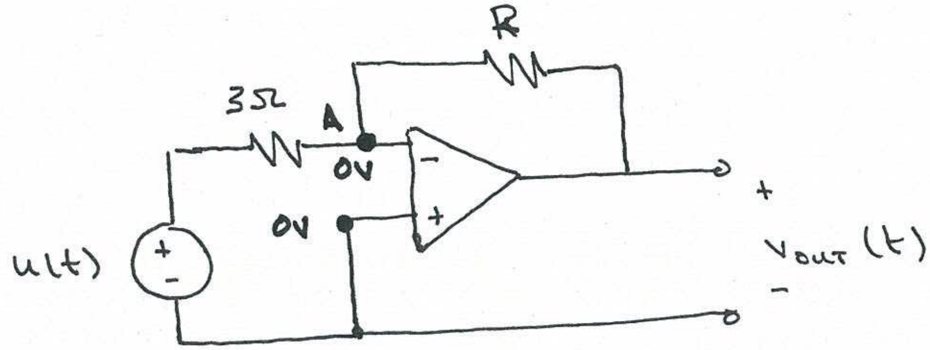
KCL at A:

$$\frac{0 - V_{in}}{8\,k\Omega} = \frac{V_{in} - V_{OUT}}{160\,k\Omega} \Rightarrow -20V_{in} = V_{in} - V_{OUT}$$

$$\boxed{V_{OUT} = 21\,V_{in}}$$

5.5

Probably easiest to find an input-output relation, and apply both inputs to that:



$$R = \frac{(8\Omega)(24\Omega)}{8\Omega + 24\Omega} = 6\Omega$$

Equivalent resistance of parallel 8Ω & 24Ω resistors

KCL at A: $\frac{u(t) - 0}{3\Omega} = \frac{0 - V_{OUT}(t)}{R} \Rightarrow 2u(t) = -V_{OUT}(t)$

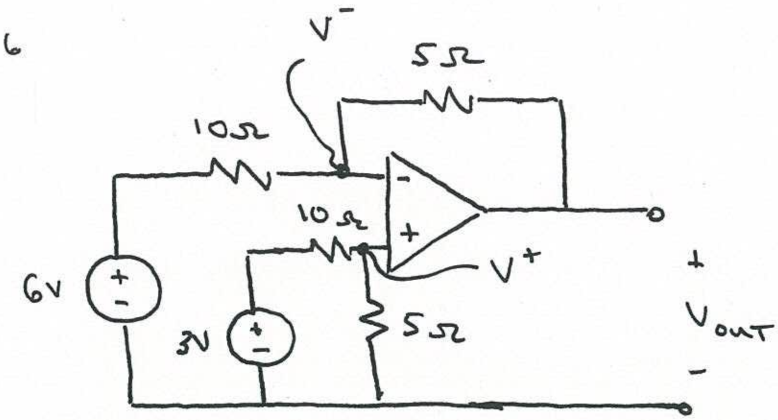
$$V_{OUT}(t) = -2u(t)$$

(a)

$V_{OUT}(t) = -2[4V] \Rightarrow \boxed{V_{OUT}(t) = -8V}$

(b) $V_{OUT}(t) = -2[2\cos(5t)] \Rightarrow \boxed{V_{OUT}(t) = -4\cos(5t)\ V}$

5.6



Non-inverting input: $\dfrac{3V - V^+}{10\,\Omega} = \dfrac{V^+ - 0V}{5\,\Omega} \Rightarrow V^+ = 1V$

Opamp rules $\Rightarrow$ $V^- = V^+$ $\Rightarrow$ $V^- = 1V$

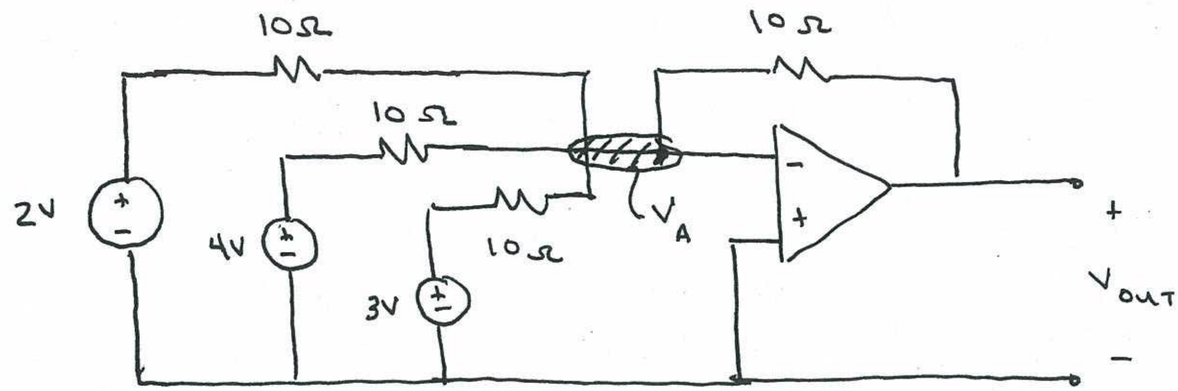
KCL at inverting input:

$$\frac{6V - V^-}{10\,\Omega} = \frac{V^- - V_{OUT}}{5\,\Omega} \longrightarrow \frac{6V - 1V}{10\,\Omega} = \frac{1V - V_{OUT}}{5\,\Omega}$$

$$5V = 2V - 2V_{OUT}$$

$$\underline{\underline{V_{OUT} = -\frac{3}{2}\,V}}$$

5.7



Op-amp rules: $V_A = 0V$

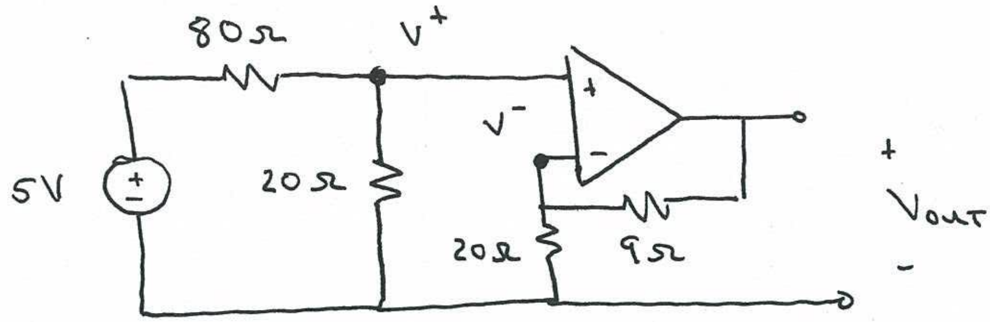
KCL at node A:

$$\frac{2V - V_A}{10\Omega} + \frac{4V - V_A}{10\Omega} + \frac{3V - V_A}{10\Omega} = \frac{V_A - V_{OUT}}{10\Omega}$$

$$\frac{2V}{10\Omega} + \frac{4V}{10\Omega} + \frac{3V}{10\Omega} = -\frac{V_{OUT}}{10\Omega}$$

$$\underline{\underline{V_{OUT} = -9V}}$$

5.8



KCL at non-inverting input:

$$\frac{5V - V^+}{80\Omega} = \frac{V^+ - 0V}{20\Omega} \Rightarrow V^+ = 1V$$

Op-amp rules: $V^- = V^+ \Rightarrow V^- = 1V$

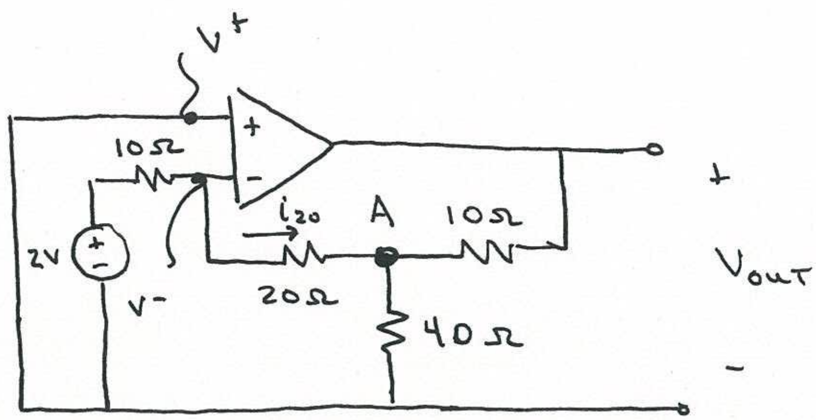
KCL at inverting input:

$$\frac{V^- - 0V}{20\Omega} + \frac{V^- - V_{OUT}}{9\Omega} = 0$$

$$\frac{1V}{20\Omega} + \frac{1V - V_{OUT}}{9\Omega} = 0$$

$$\underline{\underline{V_{OUT} = \frac{29}{20} V}}$$

5.9



$V^+ = 0 \Rightarrow \underbrace{V^- = 0}$ (op-amp rules)

KCL at A:

$$\frac{V_A - V^-}{20\Omega} + \frac{V_A - 0V}{40\Omega} + \frac{V_A - V_{OUT}}{10\Omega} = 0$$

$$\frac{V_A}{20\Omega} + \frac{V_A}{40\Omega} + \frac{V_A}{10\Omega} = \frac{V_{OUT}}{10\Omega} \Rightarrow V_{OUT} = \frac{7}{4} V_A$$

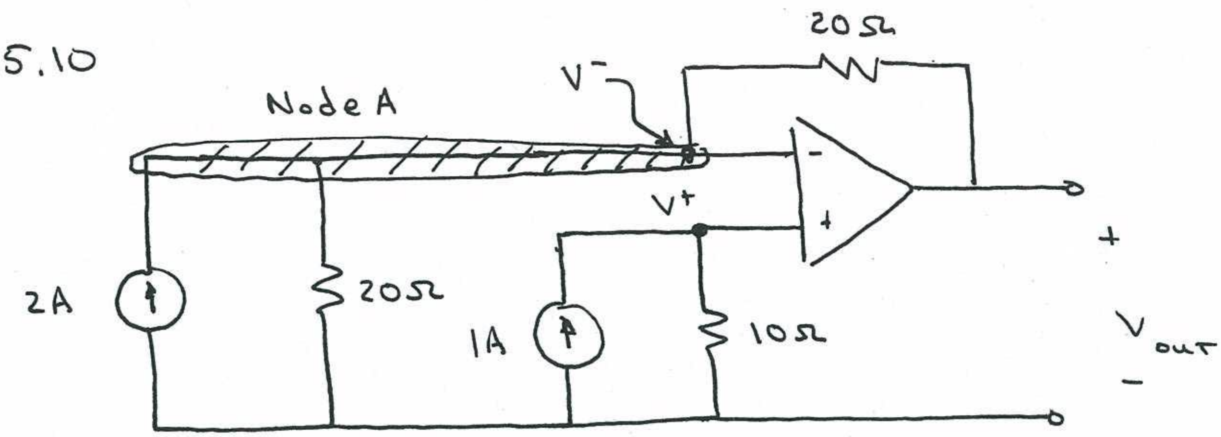
KCL at inverting input:

$$\frac{2V - V^-}{10\Omega} = i_{20} \Rightarrow i_{20} = \frac{1}{5} A$$

Ohm's Law: $V^- - V_A = (20\Omega)\, i_{20} \Rightarrow V_A = -4V$

$\underline{\underline{V_{OUT} = 7V}}$

5.10



$$V^+ = (10\,\Omega)(1A) = 10V$$

Op-amp rule: $V^+ = V^- = 10V$

KCL, node A:

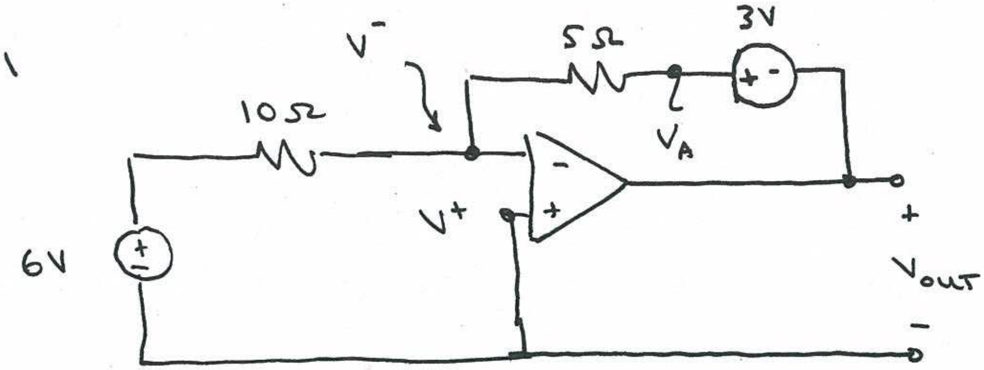
$$2A = \frac{V^-}{20\,\Omega} + \frac{V^- - V_{OUT}}{20\,\Omega}$$

$$40\,V = V^- + V^- - V_{OUT}$$

$$40V = 10V + 10V - V_{OUT}$$

$$\underline{\underline{V_{OUT} = -20V}}$$

5.11



$V^- = V^+ = 0V$ (op-amp rules)

KCL at inverting input:

$$\frac{6V - V^-}{10\,\Omega} = \frac{V^- - V_A}{5\,\Omega} \Rightarrow 6V - V^- = 2V^- - 2V_A$$

$$6V = -2V_A$$

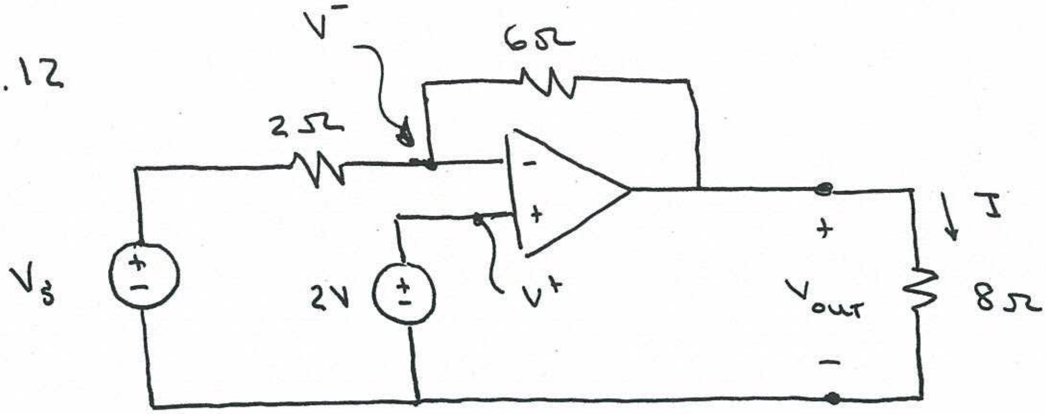
$$V_A = -3V$$

$V_A = V_{OUT} + 3V$

$$V_{OUT} = V_A - 3V$$

$$\underline{\underline{V_{OUT} = -6V}}$$

5.12



$V^+ = 2V \Rightarrow V^- = 2V$ (op-amp rules)

KCL at inverting input:

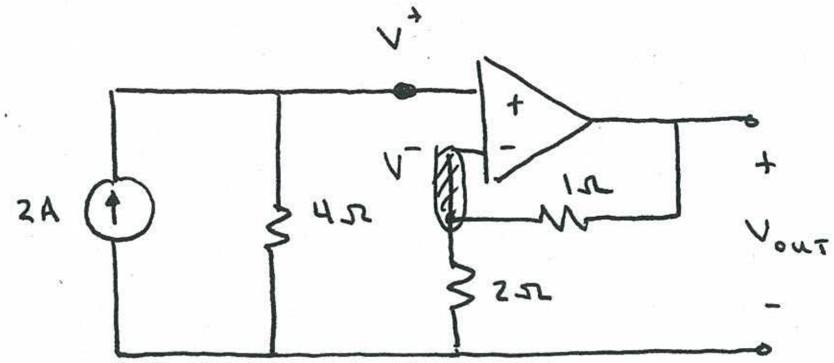
$$\frac{V_s - V^-}{2\Omega} = \frac{V^- - V_{OUT}}{6\Omega} \Rightarrow 3V_s - 4V^- = V_{OUT}$$

$$V_{OUT} = 8V - 3V_s$$

Ohm's Law:

$$I = \frac{V_{OUT}}{8\Omega} \longrightarrow \underline{\underline{I = \frac{8V - 3V_s}{8\Omega}}}$$

5.13



$V^+ = (2A)(4\Omega) = 8V$

KCL at inverting input + Ohm's law

$V^- = V^+ = 8V$ (op-amp rules)

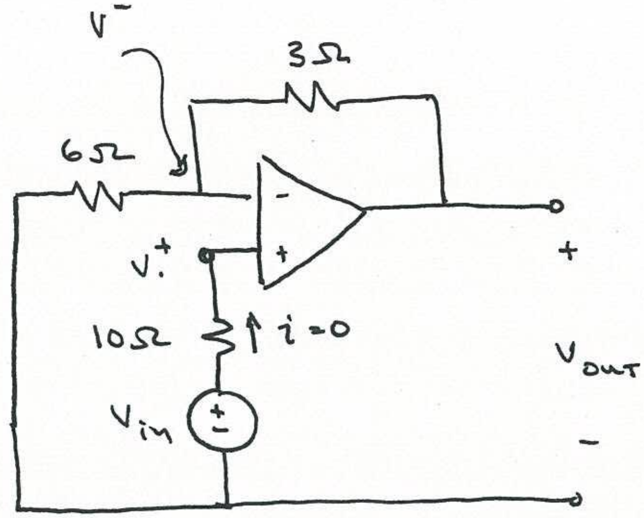
KCL at inverting input:

$$\frac{V^- - 0V}{2\Omega} + \frac{V^- - V_{OUT}}{1\Omega} = 0 \Rightarrow 3V^- = 2V_{OUT}$$

$$2V_{OUT} = (3)(8V)$$

$$\underline{\underline{V_{OUT} = 12V}}$$

5.14



$V^+ = V_{in}$

no current into input terminals, so no voltage drop across 10Ω resistor

$V^- = V^+ = V_{in}$

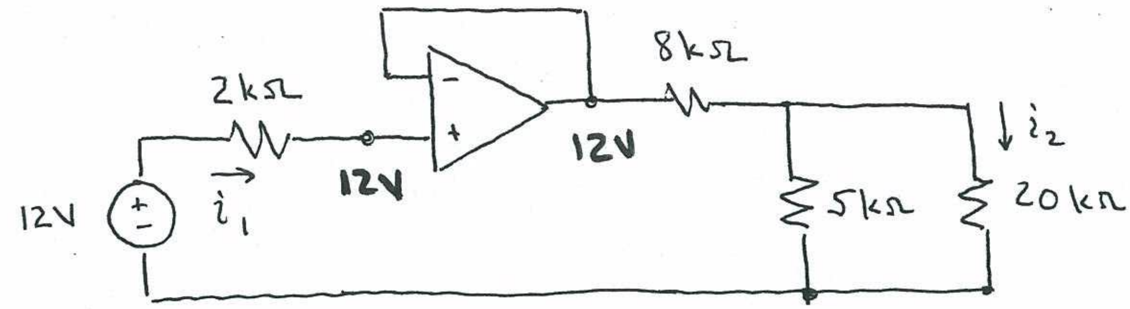
KCL at inverting input:

$$\frac{V^- - 0V}{6\Omega} = \frac{V_{OUT} - V^-}{3\Omega} \Rightarrow 2V_{OUT} = 3V^-$$

$$2V_{OUT} = 3(V_{in})$$

$$\underline{\underline{V_{OUT} = \frac{3}{2} V_{in}}}$$

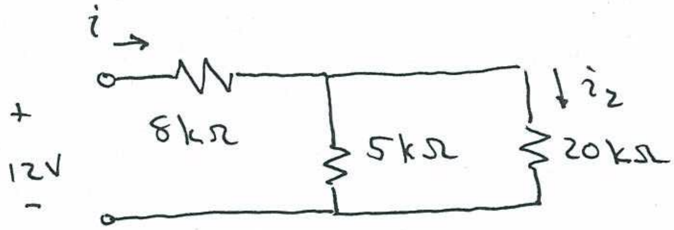
5.15



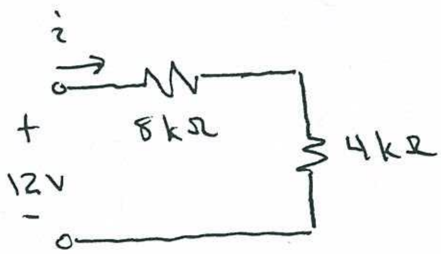
From op-amp rules, $\boxed{i_1 = 0\,\text{A}}$

From op-amp rules, op-amp output voltage = 12V,

so circuit to determine i_2 is:

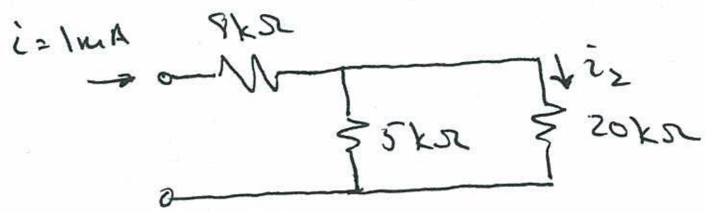


↓ Equivalent circuit



$$\Rightarrow \quad i = \frac{12\text{V}}{8\text{k}\Omega + 4\text{k}\Omega} = 1\text{mA}$$

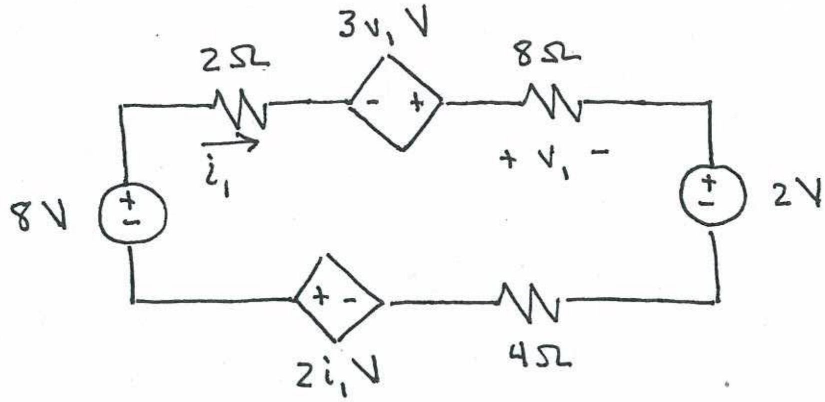
current divider:



$$i_2 = \frac{5\text{k}\Omega}{5\text{k}\Omega + 20\text{k}\Omega}(1\text{mA})$$

$$\boxed{i_2 = 0.2\text{ mA}}$$

5.16



KVL :

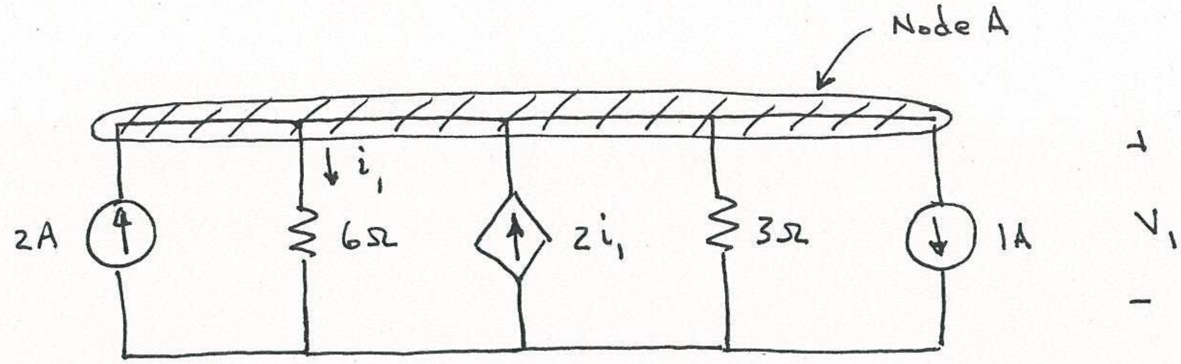
$$-8V + (2\Omega)\,i_1 - 3v_1 + v_1 + 2V + (4\Omega)\,i_1 - 2i_1 = 0$$

$$v_1 = (8\Omega)\,i_1$$

$$-6 - 2v_1 + \frac{v_1}{2} = 0 \quad \Rightarrow \quad -3v_1 = 12$$

$$\boxed{v_1 = -4V}$$

5.17



KCL, Node A:

$$-2A + i_1 - 2i_1 + \frac{V_1}{3\Omega} + 1A = 0$$

Ohm's Law:

$$i_1 = \frac{V_1}{6\Omega} \Rightarrow V_1 = 6i_1$$

$$-2 + i_1 - 2i_1 + \frac{6i_1}{3\Omega} + 1 = 0 \Rightarrow \boxed{i_1 = 1A}$$